

1. What is the role of a Certificate Authority in Hyperledger Fabric? (2 Marks)

In **Hyperledger Fabric**, a **Certificate Authority (CA)** plays a critical role in managing identities and ensuring secure communication within the network.

Role of Certificate Authority (CA):

- **Identity Management:** The CA issues digital certificates to users, peers, and organizations, giving each participant a unique identity.
- **Authentication:** It verifies the identity of participants before allowing them to access the network.
- **Authorization Support:** Certificates help define access control policies (who can read/write/invoke chaincode).
- **Secure Communication:** Enables encrypted communication using TLS certificates.
- **Certificate Lifecycle Management:** Handles registration, enrollment, renewal, and revocation of certificates.

2. Define Chaincode in Hyperledger Fabric. (2 Marks)

In **Hyperledger Fabric**, **Chaincode** is a smart contract that defines and executes the business logic of the blockchain network.

- It is a program that manages how transactions are created and processed on the ledger.
- It runs on peer nodes to validate and execute transaction proposals.
- It interacts with the ledger to read and update data (world state).
- It enforces rules and conditions agreed upon by network participants.
- It can be written in programming languages like Go, Java, and JavaScript.
- It is deployed on channels to ensure data privacy and restricted access among participants.

3. What is a Channel in Hyperledger Fabric? (2 Marks)

In **Hyperledger Fabric**, a **Channel** is a private communication layer that allows a group of participants to share data securely and separately from others in the same network.

- It is a logical partition of the blockchain network for a specific set of members.
- It ensures that only authorized participants can access the data within that channel.
- Each channel maintains its own separate ledger and transaction history.
- It provides data privacy and confidentiality among organizations.
- Smart contracts (chaincode) can be deployed on a specific channel.
- It enables multiple channels to exist within a single network for different use cases.

4. Name any two consensus mechanisms used in Hyperledger Fabric. (2 Marks)

- **Solo Consensus** – A basic consensus mechanism where a single ordering node handles all transactions. It is simple to configure and useful for learning or testing purposes, but it lacks security, scalability, and fault tolerance, so it is not suitable for production use.
- **Raft Consensus** – A leader-based distributed consensus mechanism where one node acts as a leader and others follow. It ensures consistent transaction ordering, provides crash fault tolerance, and maintains system reliability even if some nodes go offline, making it suitable for production environments.
- **Kafka Consensus** – A consensus mechanism that uses **Apache Kafka** for ordering transactions, providing high throughput and reliability, but requires additional setup and is now mostly replaced by Raft.

5. Explain the architecture of Hyperledger Fabric with its key components. (5 Marks)

The architecture of **Hyperledger Fabric** is modular and permissioned, designed mainly for enterprise use. It follows a unique **execute–order–validate** transaction process, which improves performance, security, and scalability. It uses distributed ledger technology, smart contracts (chaincode), and different types of nodes to achieve decentralized trust without using any cryptocurrency.

Key Architectural Components:

- **Peer Nodes** – These are the main nodes of the network that maintain the ledger and run chaincode. Some peers act as endorsing peers (simulate transactions), while others commit transactions to the ledger.
- **Ordering Service (Orderer)** – This component manages consensus by collecting transactions, ordering them, and forming blocks to ensure all peers have the same transaction sequence.
- **Ledger Subsystem** – It consists of two parts:
 - **Blockchain** – Stores the complete history of transactions (immutable).
 - **World State** – Stores the latest state of data (current values of assets).
- **Chaincode (Smart Contracts)** – It contains business logic that defines how transactions are executed and how data is modified in the ledger.
- **Membership Service Provider (MSP)** – It manages identities and authentication of users using digital certificates, ensuring only authorized participants can access the network.
- **Channels** – These provide privacy by allowing a subset of participants to have their own private communication and separate ledger within the network.
- **Certificate Authority (CA)** – Issues digital certificates to provide secure identity and authentication.

Transaction Flow Summary:

1. **Endorsement:** Client sends a proposal to endorsing peers.
2. **Ordering:** Endorsed transactions are sent to the ordering service.
3. **Validation & Commitment:** The orderer creates blocks and delivers them to peers for validation and committing to the ledger.

6. Describe the role of nodes, Certificate Authority, and chaincode in Hyperledger Fabric. (5 Marks)

In **Hyperledger Fabric**, nodes, Certificate Authority, and chaincode play essential roles in ensuring the network works securely, efficiently, and reliably.

Roles:

- **Nodes (Peers & Orderers)** – Nodes are the core of the network. Peer nodes maintain the ledger, execute chaincode, and validate transactions before committing them. Ordering nodes are responsible for collecting transactions from different peers, arranging them in the correct order, and creating blocks. This separation of roles improves performance and ensures consistency across the network.
- **Certificate Authority (CA)** – The CA is responsible for identity management in the network. It issues digital certificates to users, peers, and organizations, which are used for authentication and secure communication. It also supports registration, enrollment, and revocation of identities, ensuring that only trusted members can participate in the network.
- **Chaincode** – Chaincode acts as the business logic layer of the blockchain. It defines the rules and conditions under which transactions are executed. It interacts with the ledger to read and update data, ensuring that all operations follow predefined agreements. Chaincode also helps automate processes, reduce manual intervention, and maintain consistency in transaction execution.

7. Discuss the challenges of scalability and interoperability in blockchain systems. (5 Marks)

Challenges of Scalability and Interoperability in Blockchain Systems : Blockchain systems face important limitations when it comes to handling large-scale usage and working with other systems.

Scalability Challenges:

- **Limited Transaction Speed** – Many blockchains process fewer transactions per second compared to traditional systems, causing delays.
- **High Latency** – Time taken to validate and confirm transactions can be slow, especially in large networks.
- **Resource Consumption** – Consensus mechanisms require high computational power and storage, increasing cost.
- **Network Congestion** – As the number of users increases, the network can become overloaded, reducing performance.
- **Block Size & Throughput Limits** – Fixed block size and time intervals restrict the number of transactions processed.

Interoperability Challenges:

- **Lack of Standardization** – Different blockchain platforms use different protocols, making integration difficult.
- **Data Sharing Issues** – It is hard for one blockchain to access or understand data from another blockchain.
- **Cross-Chain Communication** – Secure and reliable communication between multiple blockchains is complex.
- **Compatibility Problems** – Differences in consensus mechanisms and architectures create integration barriers.
- **Security Risks** – Interoperability solutions (like bridges) may introduce vulnerabilities and increase attack risks.

8. Explain how a Hyperledger Fabric blockchain network is designed and deployed. (5 Marks)

In **Hyperledger Fabric**, designing and deploying a blockchain network involves setting up the structure, configuring components, and launching the network step by step.

Design Phase:

- **Define Network Participants** – Identify organizations, users, peers, and orderers that will be part of the network.
- **Set Policies & Permissions** – Decide access control rules, endorsement policies, and roles of each participant.
- **Design Channels** – Create channels for private communication between specific members.
- **Select Consensus Mechanism** – Choose a suitable method like Raft for transaction ordering.
- **Plan Chaincode Logic** – Define business logic and smart contracts according to application requirements.

Deployment Phase:

- **Setup Certificate Authority (CA)** – Generate and manage digital identities for all participants.
- **Configure Network Components** – Set up peers, orderers, and channels using configuration files.
- **Create Genesis Block** – Initialize the network with the first block for the ordering service.
- **Start Network Nodes** – Launch peer and orderer nodes using Docker containers or similar tools.
- **Install and Deploy Chaincode** – Install smart contracts on peers and approve them on the channel.

- **Invoke and Test Transactions** – Run transactions to verify that the network is working correctly. Bitcoin uses **cryptographic keys** to secure transactions.

9. What is an ERC20 token? Explain the steps to create an ERC20 token? (5 Marks)

An **ERC20 token** is a widely used token standard on the **Ethereum** blockchain. It defines a common set of rules and functions that all tokens must follow, such as transferring tokens, checking balances, and approving transactions. This standard ensures that all ERC20 tokens are compatible with wallets, exchanges, and decentralized applications (DApps). ERC20 tokens are commonly used for cryptocurrencies, utility tokens, and fundraising through Initial Coin Offerings (ICOs).

Steps to Create an ERC20 Token:

- **Setup Development Environment** – Install tools like **Remix IDE** or **Truffle** and a wallet like **MetaMask**.
- **Write Smart Contract** – Create a Solidity contract implementing ERC20 functions (name, symbol, total supply, transfer, etc.).
- **Compile the Contract** – Use Remix or Truffle to compile the code and check for errors.
- **Deploy the Contract** – Deploy the contract on Ethereum (testnet or mainnet) using MetaMask.
- **Verify and Interact** – After deployment, interact with the contract to transfer tokens and check balances.
- **Use or List Token** – The token can be integrated into applications, wallets, or listed on exchanges.

10. Define ERC721 token. (2 Marks)

An **ERC721 token** is a standard for creating non-fungible tokens (NFTs) on the Ethereum blockchain. Unlike ERC20 tokens (which are identical and interchangeable), ERC721 tokens are unique and non-interchangeable, meaning each token has its own distinct value and identity.

- Each token represents a unique asset such as digital art, collectibles, or game items.
- It uses a unique token ID to distinguish one token from another.
- Ownership of each token is recorded and can be transferred securely on the blockchain.
- It supports functions for transferring ownership and checking the owner of a token.
- Commonly used in NFTs, digital collectibles, gaming assets, and real-world asset tokenization.

11. What is an NFT? List its key characteristics. (2 Marks)

An NFT (Non-Fungible Token) is a unique digital asset stored on a blockchain like **Ethereum**. It represents ownership of a specific item such as digital art, music, videos, or virtual items, and cannot be replaced or exchanged equally like normal cryptocurrencies. It also ensures authenticity and prevents duplication through blockchain technology.

Key Characteristics:

- **Uniqueness** – Each NFT has a unique identity and cannot be duplicated.
- **Non-Fungibility** – NFTs are not interchangeable (one NFT ≠ another NFT).
- **Ownership Proof** – Ownership is securely recorded on the blockchain.
- **Indivisibility** – NFTs cannot be divided into smaller units like cryptocurrencies.
- **Transparency** – All transactions and ownership history are publicly verifiable.
- **Transferability** – NFTs can be bought, sold, or transferred between users.

12. What is an ICO? List its key characteristics. (2 Marks)

An **ICO (Initial Coin Offering)** is a fundraising method used by blockchain projects where new tokens are sold to investors in exchange for cryptocurrencies like Bitcoin or Ether. It is commonly used by startups to raise capital for developing new blockchain-based applications.

Key Characteristics:

- **Token-Based Fundraising** – Investors receive project tokens in return for their investment.
- **Early Investment Opportunity** – Allows investors to buy tokens at an early stage, often at a lower price.
- **Decentralized Funding** – Does not require traditional financial institutions like banks.
- **High Risk & High Return** – Investments can give high profits but also carry significant risk.
- **Global Participation** – Anyone with internet access and cryptocurrency can participate.
- **Whitepaper Requirement** – Projects usually release a whitepaper explaining their idea, technology, and roadmap.

13. Name any two types of cryptocurrencies. (2 Marks)

- **Coins** – These are native digital currencies that operate on their own blockchain, such as **Bitcoin** and **Ethereum**. They are mainly used as a medium of exchange, store of value, and for making transactions without any central authority. Coins also help in maintaining and securing their own blockchain network.
- **Tokens** – These are digital assets created on existing blockchains like **Ethereum**. They do not have their own blockchain and are used for specific purposes such as accessing services, representing assets, or enabling smart contract operations in applications.
- **Stablecoins** – These are cryptocurrencies designed to maintain a stable value by being linked to assets like fiat currency (e.g., USD). For example, **Tether** is widely used to reduce price volatility in the crypto market.

14. Compare ERC20 and ERC721 tokens with suitable examples. (5 Marks)

Basis	ERC20 Token	ERC721 Token
Definition	A standard for fungible tokens on Ethereum	A standard for non-fungible tokens (NFTs) on Ethereum
Nature	Fungible (all tokens are identical)	Non-fungible (each token is unique)
Interchangeability	Can be exchanged one-to-one	Cannot be exchanged equally
Use Case	Used for cryptocurrencies, utility tokens	Used for NFTs like digital art, gaming assets
Divisibility	Can be divided into smaller units	Cannot be divided
Ownership	Balance-based (how many tokens you own)	Ownership-based (which specific token you own)
Token Identity	No unique identity for each token	Each token has a unique ID
Transfer	Simple transfer of amount	Transfer of specific token ID
Example	Tether , Chainlink	CryptoKitties, digital artworks

15. Explain NFTs and their applications in real-world scenarios. (5 Marks)

NFTs (Non-Fungible Tokens) are unique digital assets stored on a blockchain like **Ethereum**. Each NFT has a unique identifier and metadata that distinguishes it from other tokens, making it impossible to duplicate or replace. They use smart contracts to manage ownership and transfer, ensuring secure and transparent transactions without intermediaries.

Applications of NFTs in Real-World:

- **Digital Art & Collectibles** – NFTs allow artists to tokenize their artwork and sell it globally. They ensure proof of ownership and authenticity, and creators can also earn royalties whenever their work is resold.
- **Gaming** – NFTs are used to represent in-game assets like skins, weapons, and characters. Players truly own these assets and can trade or sell them outside the game, creating real economic value.
- **Music & Entertainment** – Musicians and creators can release songs, albums, or exclusive content as NFTs. This helps them connect directly with fans and earn revenue without relying on intermediaries.
- **Real Estate** – NFTs can represent ownership of physical properties by converting them into digital tokens. This makes property transactions faster, more transparent, and reduces paperwork.
- **Identity & Certification** – Important documents like certificates, IDs, and academic records can be stored as NFTs. This ensures they are tamper-proof, verifiable, and easily accessible.
- **Supply Chain Management** – NFTs help track the origin and ownership of products. This improves transparency, prevents counterfeiting, and builds trust among consumers.
- **Ticketing & Events** – NFTs can be used as digital tickets for concerts or events. They reduce fraud, allow easy verification, and can include extra benefits like exclusive access or rewards.

16. Differentiate between ICO and STO with advantages and disadvantages. (5 Marks)

Basis	ICO (Initial Coin Offering)	STO (Security Token Offering)
Definition	A fundraising method where tokens are issued without giving ownership rights	A fundraising method where tokens represent real-world assets or securities
Regulation	Mostly unregulated	Highly regulated by government authorities
Investor Rights	No ownership or legal rights	Provides ownership, dividends, or profit rights
Risk Level	High risk due to scams and lack of control	Lower risk due to legal compliance
Transparency	Less transparent	More transparent and secure
Cost & Complexity	Easy and low-cost to launch	Expensive and complex due to legal procedures
Liquidity	High liquidity (easy to trade tokens)	Lower liquidity due to regulatory restrictions
Legal Compliance	Minimal or no legal requirements	Strict legal compliance required
Example Use	Utility tokens for apps	Tokenized shares, real estate, assets

17. What is the role of blockchain in IoT? (2 Marks)

Blockchain plays an important role in improving the security, transparency, and efficiency of IoT (Internet of Things) systems.

Role of Blockchain in IoT:

- **Enhanced Security** – Blockchain provides secure data storage using encryption, protecting IoT devices from hacking and unauthorized access.
- **Decentralization** – Removes the need for a central authority, reducing single points of

failure in IoT networks.

- **Data Integrity** – Ensures that data generated by IoT devices cannot be altered or tampered with once recorded.
- **Device Authentication** – Helps verify the identity of IoT devices, allowing only trusted devices to communicate.
- **Automation using Smart Contracts** – Enables automatic execution of actions (e.g., triggering devices) based on predefined conditions.
- **Transparency & Traceability** – All transactions and data exchanges are recorded and can be tracked easily.
- **Efficient Data Sharing** – Allows secure and reliable data sharing between multiple devices and systems.

18. How does blockchain enhance cybersecurity? (5 Marks)

Blockchain enhances cybersecurity by providing a secure, transparent, and tamper-proof way to store and manage data.

How Blockchain Enhances Cybersecurity:

- **Decentralization** – Data is stored across multiple nodes instead of a single server, reducing the risk of single-point failures and attacks, making the system more resilient.
- **Data Immutability** – Once data is recorded, it cannot be changed or deleted, preventing tampering and unauthorized modifications, ensuring data integrity.
- **Strong Encryption** – Uses cryptographic techniques to secure data and transactions from unauthorized access, protecting sensitive information.
- **Secure Identity Management** – Helps in verifying identities using digital signatures, reducing identity theft and fraud, and ensuring trusted access.
- **Protection Against DDoS Attacks** – Decentralized structure makes it harder for attackers to target and shut down the network, improving availability.
- **Transparency & Auditability** – All transactions are recorded and can be verified, making it easier to detect suspicious activities and track issues.
- **Secure Data Sharing** – Enables safe and controlled sharing of data between parties without intermediaries, increasing trust and efficiency.
- **Consensus Mechanisms** – Transactions are validated through consensus, ensuring only legitimate data is added and preventing unauthorized changes.

19. What is a smart contract in IoT-based blockchain systems? (2 Marks)

A smart contract in IoT-based blockchain systems is a self-executing program stored on a blockchain like Ethereum that automatically performs actions when certain conditions are met between IoT devices.

Explanation:

- It defines rules and conditions for how IoT devices should interact with each other.
- It automatically executes actions (e.g., turning on a device, sending data) without human intervention.
- It ensures secure and trusted communication between devices using blockchain.
- It reduces the need for intermediaries, making processes faster and more efficient.
- It records all actions on the blockchain, ensuring transparency and traceability.
- It minimizes human errors by automating decision-making processes.
- It improves system efficiency by enabling real-time responses between IoT devices.

20. Describe the applications of blockchain in Artificial Intelligence. (2 Marks)

Blockchain has important applications in **Artificial Intelligence (AI)** by improving data security, transparency, and trust in AI systems.

Applications of Blockchain in AI:

- **Secure Data Sharing** – Blockchain enables safe and tamper-proof sharing of data used to train AI models, improving data reliability.
- **Data Integrity** – Ensures that the data used in AI systems is accurate and has not been altered, leading to better decision-making.
- **Decentralized AI Models** – AI models can be trained and operated in a decentralized manner, reducing dependency on central authorities.
- **Transparency in AI Decisions** – Blockchain records AI processes, making decisions more explainable and traceable.
- **Automation using Smart Contracts** – Smart contracts can automate AI workflows, such as triggering actions based on AI predictions.
- **Secure Data Marketplaces** – Allows users to buy and sell data for AI training securely while maintaining ownership rights.
- **Trust & Accountability** – Blockchain provides a clear record of data and actions, increasing trust in AI systems.